

PDE Work Roadmap for a Supersonic Inlet CFD Project

From Conservation Laws to 2D Compressible Flow

Fall 2026 Preparation

1 Purpose

The goal is not to jump immediately into a CFD package and generate contour plots. The goal is to understand the partial differential equations that a compressible-flow solver is approximating, then build toward the governing equations of a supersonic inlet.

The recommended progression is

Advection → Burgers → Potential Flow → 1D Euler → Shock Tube → Quasi-1D Nozzle → 2D Euler → Supersonic Inlet
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The full compressible Navier–Stokes equations come after this foundation.

2 Recommended PDE Progression

Level	PDE/System	Main lesson	Inlet relevance
1	1D linear advection	Characteristics, CFL stability, numerical propagation	High
2	Inviscid Burgers equation	Nonlinearity, shocks, weak solutions	Very high
3	Laplace equation	Potential flow, elliptic PDEs, boundary conditions	Moderate
4	1D Euler equations	Compressible waves, conservation laws, shocks	Essential
5	Quasi-1D Euler equations	Choking, area variation, internal shocks	Essential
6	2D Euler equations	Oblique shocks and inviscid inlet flow	Essential
7	Compressible Navier–Stokes	Viscosity, heat transfer, boundary layers	Final stage

3 Problem 1: Linear Advection

Begin with

$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} = 0$$

Use the method of characteristics to show

$$u(x, t) = f(x - at).$$

Required work

1. Derive the characteristic equation $dx/dt = a$.
2. Explain physically why the waveform translates without changing shape.
3. Use the initial condition

$$u(x, 0) = \begin{cases} 1, & 0.25 < x < 0.5, \\ 0, & \text{otherwise.} \end{cases}$$

4. Implement first-order upwinding:

$$u_i^{n+1} = u_i^n - C(u_i^n - u_{i-1}^n), \quad C = \frac{a\Delta t}{\Delta x}.$$

5. Run $C = 0.5, 0.9, 1.0, 1.1$.
6. Explain numerical diffusion and why the $C > 1$ case becomes unstable.

Do not move on until you understand: characteristics, domain of dependence, CFL stability, truncation error, and numerical diffusion.

4 Problem 2: Inviscid Burgers Equation

Study

$$\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = 0$$

and its conservation form

$$\frac{\partial u}{\partial t} + \frac{\partial}{\partial x} \left(\frac{u^2}{2} \right) = 0.$$

Characteristics satisfy

$$\frac{dx}{dt} = u.$$

Unlike linear advection, the propagation speed depends on the solution itself.

Shock problem

Use

$$u(x, 0) = \begin{cases} 2, & x < 0, \\ 1, & x > 0. \end{cases}$$

Derive the Rankine–Hugoniot condition

$$s = \frac{f(u_R) - f(u_L)}{u_R - u_L}$$

for $f(u) = u^2/2$, then calculate the shock speed.

Rarefaction problem

Reverse the states:

$$u_L = 1, \quad u_R = 2.$$

Derive the self-similar rarefaction solution and explain why the characteristics spread rather than intersect.

Required concepts

- Nonlinear hyperbolic PDEs
- Conservation form
- Characteristic crossing
- Shock formation
- Weak solutions
- Rankine–Hugoniot conditions
- Entropy condition

5 Problem 3: Laplace Equation and Potential Flow

Solve the elliptic PDE

$$\boxed{\nabla^2 \phi = 0}$$

or in two dimensions

$$\boxed{\frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} = 0.}$$

Define

$$u = \frac{\partial \phi}{\partial x}, \quad v = \frac{\partial \phi}{\partial y}.$$

Required work

1. Derive Laplace’s equation from incompressibility and irrotationality.
2. Solve at least one rectangular-domain boundary-value problem analytically by separation of variables.
3. Write a finite-difference Laplace solver.
4. Solve a potential-flow problem around a simple body or imposed boundary.
5. Plot equipotential lines and streamlines.
6. Explain why potential flow cannot model viscous boundary layers or shocks.

This stage forces you to work with an elliptic PDE before returning to hyperbolic compressible-flow systems.

6 Problem 4: One-Dimensional Compressible Euler Equations

The inviscid compressible conservation laws are

Mass

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u)}{\partial x} = 0$$

Momentum

$$\frac{\partial(\rho u)}{\partial t} + \frac{\partial(\rho u^2 + p)}{\partial x} = 0$$

Energy

$$\frac{\partial E}{\partial t} + \frac{\partial[u(E + p)]}{\partial x} = 0$$

with

$$E = \frac{p}{\gamma - 1} + \frac{1}{2}\rho u^2.$$

Write the system as

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}}{\partial x} = 0$$

where

$$\mathbf{U} = \begin{bmatrix} \rho \\ \rho u \\ E \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ u(E + p) \end{bmatrix}.$$

Required derivations

1. Derive mass conservation from a differential control volume.
2. Derive momentum conservation.
3. Derive total-energy conservation.
4. Derive the ideal-gas closure relation in terms of conservative variables.
5. Calculate the flux Jacobian

$$\mathbf{A} = \frac{\partial \mathbf{F}}{\partial \mathbf{U}}.$$

6. Show that its characteristic speeds are

$$\lambda_1 = u - a, \quad \lambda_2 = u, \quad \lambda_3 = u + a.$$

7. Explain what each characteristic speed means physically.

Do not treat the eigenvalues as abstract linear algebra. They describe how information propagates through compressible flow.

7 Problem 5: Sod Shock Tube

Use the Euler equations to solve the Riemann problem

$$(\rho, u, p)_L = (1, 0, 1), \quad (\rho, u, p)_R = (0.125, 0, 0.1).$$

The solution contains

rarefaction $\rightarrow$ contact discontinuity $\rightarrow$ shock.

Required outputs

Plot

$$\rho(x), \quad u(x), \quad p(x), \quad M(x).$$

Compare your numerical solution against an exact or high-quality reference solution.

Numerical methods to learn

At minimum, understand:

- Finite-volume discretization
- Cell averages
- Numerical fluxes
- CFL condition for Euler equations
- First-order Godunov-type ideas
- Rusanov/local Lax–Friedrichs flux

After that, investigate Roe, HLL, or HLLC fluxes.

8 Problem 6: Quasi-One-Dimensional Euler Nozzle

Introduce a varying cross-sectional area $A(x)$:

$$\frac{\partial(A\rho)}{\partial t} + \frac{\partial(A\rho u)}{\partial x} = 0$$

$$\frac{\partial(A\rho u)}{\partial t} + \frac{\partial[A(\rho u^2 + p)]}{\partial x} = p \frac{dA}{dx}$$

$$\frac{\partial(AE)}{\partial t} + \frac{\partial[Au(E + p)]}{\partial x} = 0.$$

Analytical benchmark

Derive or carefully reproduce the area–Mach relation

$$\frac{A}{A^*} = \frac{1}{M} \left[\frac{2}{\gamma + 1} \left(1 + \frac{\gamma - 1}{2} M^2 \right) \right]^{\frac{\gamma + 1}{2(\gamma - 1)}}.$$

Required simulations

Vary back pressure until you obtain:

1. Entirely subsonic flow
2. Choked flow
3. Supersonic acceleration downstream of the throat
4. A normal shock inside the nozzle
5. A fully supersonic exit solution

Compare numerical Mach number and pressure against analytical compressible-flow relations wherever possible.

9 Problem 7: Two-Dimensional Euler Equations

This is the central PDE system for an inviscid supersonic-inlet model:

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{\partial \mathbf{F}}{\partial x} + \frac{\partial \mathbf{G}}{\partial y} = 0.$$

Use

$$\mathbf{U} = \begin{bmatrix} \rho \\ \rho u \\ \rho v \\ E \end{bmatrix},$$

$$\mathbf{F} = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ u(E + p) \end{bmatrix}, \quad \mathbf{G} = \begin{bmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ v(E + p) \end{bmatrix},$$

with

$$E = \frac{p}{\gamma - 1} + \frac{1}{2}\rho(u^2 + v^2).$$

10 Verification Case: Supersonic Flow Over a Wedge

Before attempting an inlet, solve a wedge problem such as

$$M_\infty = 2.0, \quad \theta = 10^\circ, \quad \gamma = 1.4.$$

Your numerical solution should produce an attached oblique shock.

Compare the measured numerical shock angle against the analytical $\theta - \beta - M$ relation:

$$\tan \theta = 2 \cot \beta \frac{M_\infty^2 \sin^2 \beta - 1}{M_\infty^2 (\gamma + \cos 2\beta) + 2}.$$

Report

$$\beta_{\text{CFD}}, \quad \beta_{\text{theory}}, \quad \% \text{ error}.$$

Also compare pressure ratio and downstream Mach number with oblique-shock theory.
This verification case should be completed before trusting the inlet results.

11 Problem 8: Supersonic Inlet

Once the wedge solver is verified, replace the wedge geometry with the inlet geometry.
Suggested freestream sweep:

$$M_\infty = 1.5, 2.0, 2.5, 3.0, 4.0.$$

Vary ramp angle, contraction ratio, or other geometric parameters.

Quantities to examine

$$\frac{p}{p_\infty}, \quad \frac{\rho}{\rho_\infty}, \quad M(x, y), \quad \frac{p_0}{p_{0,\infty}}.$$

A useful inlet performance metric is total-pressure recovery:

$$\pi_d = \frac{p_{0,\text{exit}}}{p_{0,\infty}}.$$

The engineering question is no longer simply “can CFD produce a shock?” Instead ask:

How does inlet geometry trade compression against shock losses and total-pressure recovery?

Investigate shock placement, shock interactions, compression ratio, exit-flow uniformity, and sensitivity to freestream Mach number.

12 Problem 9: Compressible Navier–Stokes

The eventual governing equations are schematically

$$\frac{\partial \mathbf{U}}{\partial t} + \nabla \cdot \mathbf{F}_c = \nabla \cdot \mathbf{F}_v.$$

For a Newtonian fluid,

$$\tau_{ij} = \mu \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} - \frac{2}{3} \delta_{ij} \nabla \cdot \mathbf{u} \right),$$

and Fourier heat conduction gives

$$\mathbf{q} = -k \nabla T.$$

This introduces the physics required for

shock + boundary layer + separation + heat transfer.

Do not begin by writing a full production Navier–Stokes solver. First derive the equations and understand what the viscous terms add. An established CFD package can then be used to investigate shock–boundary-layer interaction after the Euler work is verified.

13 Mathematics You Are Expected to Learn

If any of the following are unfamiliar, look them up and work through them rather than skipping them:

- Hyperbolic, elliptic, and parabolic PDE classification
- Method of characteristics
- Conservation laws
- Integral and differential conservation equations
- Weak solutions
- Rankine–Hugoniot jump conditions
- Eigenvalues and eigenvectors of flux Jacobians
- Characteristic variables
- Riemann problems
- Finite-volume methods
- Numerical flux functions
- CFL stability
- Artificial/numerical diffusion
- Consistency, stability, and convergence
- Boundary conditions for subsonic and supersonic flow

14 Suggested Repository Structure

```
supersonic-inlet-cfd/
|-- src/
|   |-- advection_1d.py
|   |-- burgers_1d.py
|   |-- potential_flow_2d.py
|   |-- euler_1d.py
|   |-- sod_shock_tube.py
|   |-- nozzle_q1d.py
|   |-- euler_2d.py
|   |-- supersonic_wedge.py
|   '-- inlet_solver.py
|
|-- docs/
|   |-- governing_equations.pdf
|   |-- numerical_methods.pdf
|   '-- verification.pdf
|
|-- results/
|   |-- shock_tube/
|   |-- nozzle/
|   |-- wedge/
|   '-- inlet/
|
|-- data/
|-- geometry/
'-- README.md
```


15 Minimum Standard Before Calling the Project Complete

You should be able to do all of the following without hand-waving:

1. Derive the compressible Euler equations from conservation principles.
2. Explain why they form a hyperbolic system.
3. Explain the physical meaning of $u - a$, u , and $u + a$.
4. Explain why shocks require weak solutions.
5. Derive the Rankine–Hugoniot condition.
6. Explain why a finite-volume method is natural for conservation laws.
7. Solve and verify a shock-tube problem.
8. Model choking and shocks in a quasi-1D nozzle.
9. Predict an oblique-shock angle analytically.
10. Recover approximately the same shock angle numerically.
11. Run the inlet only after the simpler solver has been verified.
12. Report numerical error, grid sensitivity, and conservation behavior rather than only contour plots.

16 Bottom Line

For this project, the most valuable progression is not learning a CFD graphical interface first. It is learning how conservation laws become nonlinear hyperbolic PDEs and how finite-volume methods approximate those equations while correctly handling discontinuities.

The 2D compressible Euler equations should be the main mathematical target. The advection, Burgers, shock-tube, nozzle, and wedge problems are the training and verification cases that make the final inlet simulation defensible.